

Quantum Computing and Quantum Machine Learning

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Plans for the week of March 2-6

1. Reminder on basics of the VQE method and how to perform measurements for the simpler one- and two-qubit Hamiltonians
2. Simulating efficiently Hamiltonians on quantum computers with the VQE method and gradient descent to optimize the state function ansatz
3. Introducing the Lipkin model
4. Work on project 1

Readings

1. For the discussion of one-qubit, two-qubit and other gates, sections 2.6-2.11 and 3.1-3.4 of Hundt's book **Quantum Computing for Programmers**, contain most of the relevant information.
2. The VQE algorithm is discussed in Hundt's section 6.11, note that the solution of the two-qubit system is outdated, measuring on one qubit only is not the present standard
3. See also the VQE review article by Tilly et al.
4. Jupyter-notebook on VQE and single-qubit problem at https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/pub/week6/ipynb/single_qubit_vqe_modified.ipynb
5. Jupyter-notebook on VQE and two-qubit problem at https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/pub/week6/ipynb/two_qubit_vqe_modified.ipynb

More background material

For the Lipkin model, we recommend strongly the work of LaRose and collaborators, see <https://journals.aps.org/prc/abstract/10.1103/PhysRevC.106.024319>, the article is also available at <https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/articles/PhysRevC.106.024319.pdf>. See in particular section 3.

Two-qubit Hamiltonian

Here we discuss how to rewrite the two-qubit Hamiltonian defined by the following Hamiltonian matrix (project 1)

$$\mathcal{H} = \begin{bmatrix} \epsilon_1 + V_z & 0 & 0 & V_x \\ 0 & \epsilon_2 - V_z & V_x & 0 \\ 0 & H_x & \epsilon_3 - V_z & 0 \\ H_x & 0 & 0 & \epsilon_4 + V_z \end{bmatrix}.$$

This Hamiltonian can be rewritten in terms of various one-qubit matrices.

Definitions

We define

$$\epsilon_{II} = \frac{\epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4}{4},$$

$$\epsilon_{ZI} = \frac{\epsilon_1 + \epsilon_2 - \epsilon_3 - \epsilon_4}{4},$$

$$\epsilon_{IZ} = \frac{\epsilon_1 - \epsilon_2 + \epsilon_3 - \epsilon_4}{4},$$

$$\epsilon_{ZZ} = \frac{\epsilon_1 - \epsilon_2 - \epsilon_3 + \epsilon_4}{4}.$$

The Hamiltonian in terms of Pauli-**X** and Pauli-**Z** matrices

With these definitions we can rewrite our two-qubit Hamiltonian as

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_I$$

with

$$\mathcal{H}_0 = \epsilon_{II} I \otimes I + \epsilon_{ZI} Z \otimes I + \epsilon_{IZ} I \otimes Z + \epsilon_{ZZ} Z \otimes Z,$$

and

$$\mathcal{H}_I = V_z Z \otimes Z + V_x X \otimes X.$$

Lipkin model

We will study a schematic model (the Lipkin model, see Nuclear Physics **62** (1965) 188), for the interaction among 2 and more fermions that can occupy two different energy levels.

In project 1 we consider a two-fermion case and a four-fermion case. For four fermions, the case we consider in the examples here, each level has degeneration $d = 4$, leading to different total spin values. The two levels have quantum numbers $\sigma = \pm 1$, with the upper level having $2\sigma = +1$ and energy $\varepsilon_1 = \varepsilon/2$. The lower level has $2\sigma = -1$ and energy $\varepsilon_2 = -\varepsilon/2$. That is, the lowest single-particle level has negative spin projection (or spin down), while the upper level has spin up. In addition, the substates of each level are characterized by the quantum numbers $p = 1, 2, 3, 4$.

Four fermion case

We define the single-particle states (for the four fermion case which we will work on here)

$$|u_{\sigma=-1,p}\rangle = a_{-p}^{\dagger}|0\rangle \quad |u_{\sigma=1,p}\rangle = a_{+p}^{\dagger}|0\rangle.$$

The single-particle states span an orthonormal basis.

Hamiltonian

The Hamiltonian of the system is given by

$$\hat{H} = \hat{H}_0 + \hat{H}_1 + \hat{H}_2$$

$$\hat{H}_0 = \frac{1}{2}\varepsilon \sum_{\sigma,p} \sigma a_{\sigma,p}^\dagger a_{\sigma,p}$$

$$\hat{H}_1 = \frac{1}{2}V \sum_{\sigma,p,p'} a_{\sigma,p}^\dagger a_{\sigma,p'}^\dagger a_{-\sigma,p'} a_{-\sigma,p}$$

$$\hat{H}_2 = \frac{1}{2}W \sum_{\sigma,p,p'} a_{\sigma,p}^\dagger a_{-\sigma,p'}^\dagger a_{\sigma,p'} a_{-\sigma,p}$$

where V and W are constants. The operator H_1 can move pairs of fermions while H_2 is a spin-exchange term. The latter moves a pair of fermions from a state $(p\sigma, p' - \sigma)$ to a state $(p - \sigma, p'\sigma)$.

Quasispin operators

We are going to rewrite the above Hamiltonian in terms of so-called quasispin operators

$$\hat{J}_+ = \sum_p a_{p+}^\dagger a_{p-}$$

$$\hat{J}_- = \sum_p a_{p-}^\dagger a_{p+}$$

$$\hat{J}_z = \frac{1}{2} \sum_{p\sigma} \sigma a_{p\sigma}^\dagger a_{p\sigma}$$

$$\hat{J}^2 = J_+ J_- + J_z^2 - J_z$$

These operators obey the commutation relations for angular momentum.

Including the number operator

We can in turn express $\hat{H}$ in terms of the above quasispin operators and the number operator

$$\hat{N} = \sum_{p\sigma} a_{p\sigma}^\dagger a_{p\sigma}.$$

Final Hamiltonian

Using the quasispin operators we can rewrite the Hamiltonian as

$$H = \varepsilon J_z + \frac{1}{2} V (J_+^2 + J_-^2) + \frac{1}{2} W (-N + J_+ J_- + J_- J_+). \quad (1)$$

Original Lipkin Hamiltonian

The Lipkin model Hamiltonian in terms of quasispin operators reads

$$H = \varepsilon J_z + \frac{1}{2} V (J_+^2 + J_-^2) + \frac{1}{2} W (-N + J_+ J_- + J_- J_+).$$

We use the standard commutation relations

$$[J_+, J_-] = 2J_z,$$

and the Casimir operator

$$J^2 = J_z^2 + \frac{1}{2}(J_+ J_- + J_- J_+).$$

More compact expression

From this we obtain

$$J_+ J_- = J^2 - J_z(J_z - 1), \quad J_- J_+ = J^2 - J_z(J_z + 1).$$

Adding the two expressions:

$$J_+ J_- + J_- J_+ = 2(J^2 - J_z^2).$$

Simplified Hamiltonian

Using

$$J_+ J_- + J_- J_+ = 2(J^2 - J_z^2)$$

and insert into the Hamiltonian:

$$H = \varepsilon J_z + \frac{1}{2} V (J_+^2 + J_-^2) + \frac{1}{2} W (-N + 2(J^2 - J_z^2)) .$$

This gives

$$H = \varepsilon J_z + \frac{V}{2} (J_+^2 + J_-^2) + W (J^2 - J_z^2) - \frac{WN}{2} .$$

Physical interpretation

- ▶ The Hamiltonian is now expressed in terms of:
 - ▶ Linear term: εJ_z
 - ▶ Pairing term: $J_+^2 + J_-^2$
 - ▶ Quadratic term: J_z^2
 - *Casimir term: J^2
- ▶ The constant $-\frac{WN}{2}$ only shifts the spectrum.

Fixed quasispin sector ($j = N/2$)

In the Lipkin model we typically work in a fixed irreducible representation:

$$J^2 = j(j+1), \quad j = \frac{N}{2}.$$

Thus

$$H = \varepsilon J_z + \frac{V}{2}(J_+^2 + J_-^2) - WJ_z^2 + Wj(j+1) - \frac{WN}{2}.$$

Final simplified form

Using $N = 2j$, the constant simplifies:

$$Wj(j+1) - \frac{WN}{2} = Wj^2.$$

Final result

$$H = \varepsilon J_z + \frac{V}{2}(J_+^2 + J_-^2) - WJ_z^2 + Wj^2.$$

The term Wj^2 is a constant shift and can often be omitted.

Final Hamiltonian matrix

These five states can in turn be used as computational basis states in order to define the Hamiltonian matrix to be diagonalized. The matrix elements are given by $\langle J, J_z | H | J', J'_z \rangle$. The Hamiltonian is hermitian and we obtain after all this labor of ours

$$H_{J=2} = \begin{bmatrix} -2\varepsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\varepsilon + 3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \varepsilon + 3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\varepsilon \end{bmatrix} \quad (2)$$

Quantum Circuit, rewriting the Lipkin model in terms of Pauli matrices

To solve the Lipkin model on a quantum computer we have to solve Schrodinger's equation. To achieve this, we will use the Variational Quantum Eigensolver (VQE) discussed earlier.

Before we proceed however, we need to rewrite the quasispin operators in terms of Pauli spin matrices/operators.

If we just keep the term with the interaction V , we have

$$H = \epsilon J_z + \frac{1}{2} V (J_+^2 + J_-^2). \quad (3)$$

We have the quasispin operators

$$J_z = \sum_n j_z^{(n)},$$

and

$$J_{\pm} = \sum_n j_{\pm}^{(n)},$$

where we have the onebody operators

$$j_z^{(n)} = \frac{1}{2} \sum_{\sigma} \sigma a_{n\sigma}^{\dagger} a_{n\sigma},$$

Commutation relations

These operators obey the $SU(2)$ commutation relations

$$[J_+, J_-] = 2J_z,$$

and

$$[J_z, J_{\pm}] = \pm J_{\pm}.$$

Here the ladder operators are defined as $J_{\pm} = J_x \pm iJ_y$. With this rewriting, we can see that the total spin operator J^2 , which is defined as

$$J^2 = J_x^2 + J_y^2 + J_z^2 = \frac{1}{2}\{J_+, J_-\} + J_z^2,$$

commutes with the Hamiltonian since the Hamiltonian.

Rotation operator

We note also that the rotation operator

$$R = e^{i\phi J_z},$$

commutes with the Hamiltonian, which can be explained as follows. Writing J_z as

$$J_z = \frac{1}{2}(N_+ - N_-),$$

where $N_{\pm} = \sum_{n\pm} a_{n\pm}^{\dagger} a_{n\pm}$, allows us to see that it measures half the difference between the number of particles in the upper and lower levels.

Possible eigenvalues

Thus, the possible eigenvalues r of the signature operator are

$$r = +1, j_z = 2n \quad (4)$$

$$r = +i, j_z = 2n + \frac{1}{2} \quad (5)$$

$$r = -1, j_z = 2n + 1 \quad (6)$$

$$r = -i, j_z = 2n + \frac{3}{2} \quad (7)$$

$$(8)$$

for $n \in \mathbb{Z}$. Note that r is real or imaginary if the number of particles N is even or odd, respectively. Since, as discussed above, the Lipkin Hamiltonian conserves N , r cannot jump between being real and imaginary. Additionally, because particles must be moved in pairs, and J_z measures half the difference between particles in the upper and lower levels, j_z can only change by as

$$j_z \rightarrow \frac{1}{2}[(N_+ \pm 2n) - (N_- \mp 2n)]$$

or $j_z \rightarrow J_z \pm 2n$.

Solving on a quantum computer

To solve the Lipkin model with a quantum computer, the first step is to map the system to a set of qubits. We will restrict ourselves here to the half-filled case where the number of particles N equals the degeneracy of the states Ω . One could assign each possible state (n, σ) a qubit such that the qubit being in the state $|1\rangle$ or $|0\rangle$ would imply that the state (n, σ) is occupied or unoccupied, respectively. This mapping scheme (which we will call occupation mapping) requires 2Ω qubits.

Number of qubits

However, because there are only two energy levels in the Lipkin model, any other natural mapping is possible. In this mapping scheme (which we will call level mapping) each doublet $((n, +1), (n, -1))$ would be assigned a qubit such that the qubit being in the state $|0\rangle$ or $|1\rangle$ would imply that the particle is in the $(n, +1)$ or $(n, -1)$ state, respectively. Note that these are the only two possible configurations of the doublet as we are restricting ourselves to the half-filled case and the Lipkin Hamiltonian only moves particles between energy levels, not degenerate states. Thus the level mapping only requires Ω qubits which is half that of the occupation mapping.

Converting to Pauli matrices

The Hamiltonian takes the form

$$H = \epsilon J_z + \frac{1}{2} V (J_+^2 + J_-^2). \quad (9)$$

Plugging the mapping from the total J operators to the individual one-body j operators yields

$$H = \epsilon \sum_n j_z^{(n)} + \frac{1}{2} V \left[\left(\sum_n j_+^{(n)} \right)^2 + \left(\sum_n j_-^{(n)} \right)^2 \right] \quad (10)$$

$$= \epsilon \sum_n j_z^{(n)} + \frac{1}{2} V \sum_{n,m} \left(j_+^{(n)} j_+^{(m)} + j_-^{(n)} j_-^{(m)} \right) \quad (11)$$

$$= \epsilon \sum_n j_z^{(n)} + 2V \sum_{n < m} \left(j_x^{(n)} j_x^{(m)} - j_y^{(n)} j_y^{(m)} \right), \quad (12)$$

where we have used the definitions

$$j_{\pm}^{(n)} = j_x^{(n)} \pm i j_y^{(n)}.$$

Explicit transformations

To convert to Pauli matrices, we make the transformations

$$j_x^{(n)} \rightarrow X_n/2,$$

and

$$j_y^{(n)} \rightarrow Y_n/2,$$

and finally

$$j_z^{(n)} \rightarrow Z_n/2,$$

which preserve the above $SU(2)$ commutation relations. The factor of $1/2$ is due to the eigenvalues of the Pauli matrices being ± 1 while we are dealing with spin $1/2$ particles.

Hamiltonian in terms of Pauli matrices

This transforms our Hamiltonian into

$$H = \frac{1}{2}\epsilon \sum_{k=1}^n Z_k + \frac{1}{2}V \sum_{n \neq j=1}^N (X_k X_j - Y_k Y_j).$$

With this form, we can clearly see that the first (one-body) term in the Hamiltonian returns the energy $-\epsilon/2$ or $+\epsilon/2$ if the qubit representing the particle of a doublet is in the ground ($|1\rangle$) or excited ($|0\rangle$) state, respectively. The action of the second (two-body) term in the Hamiltonian can be determined by noting that

$$\frac{1}{2}(XX - YY)|00\rangle = |11\rangle, \quad (13)$$

$$\frac{1}{2}(XX - YY)|01\rangle = 0, \quad (14)$$

$$\frac{1}{2}(XX - YY)|10\rangle = 0, \quad (15)$$

$$\frac{1}{2}(XX - YY)|11\rangle = |00\rangle. \quad (16)$$

That is, the two body term moves a pair of particles between the

Summarizing the Lipkin model

The Lipkin model is rewritten in terms of the quasipin operators as

$$H_0 = \epsilon J_z,$$

$$H_1 = \frac{1}{2} V (J_+^2 + J_-^2),$$

$$H_2 = \frac{1}{2} W (J_+ J_- + J_- J_+ - N),$$

Quasispin operators

These quasi spin operators obey the normal spin commutator relations

$$\begin{aligned} [J_z, J_{\pm}] &= \pm J_{\pm}, & [J_+, J_-] &= 2J_z, \\ [J^2, J_{\pm}] &= 0, & [J^2, J_z] &= 0, \end{aligned}$$

in addition to commuting with the number operator

$$[N, J_z] = [N, J_{\pm}] = [N, J^2] = 0.$$

Using these relations we can show that the Hamiltonian (a product of quasi spin operators and the number operator) also commutes with J^2 , that is $[H, J^2] = 0$. This means that H and J^2 have a shared eigenbasis, with J being a so-called **good** quantum number.

System to diagonalize by traditional methods

Using spin-eigenstates as the Hamiltonian basis, we define states through the normal approach $|J, J_z\rangle$ with J and J_z as spin and spin-projections, respectively. The states $J_z = \pm J$ are the easiest to construct, corresponding to a single level being completely filled. States in between can then be found using the quasi-spin ladder operators following

$$J_{\pm}|J, J_z\rangle = \sqrt{J(J+1) - J_z(J_z \pm 1)}|J, J_z \pm 1\rangle.$$

Using this basis for the quasi-spin Hamiltonian, we can rewrite the explicit matrix $H_{J_z, J'_z} = \langle J, J_z | H | J, J'_z \rangle$ can be constructed.

The $J = 1$ case

For $N = 2$ particles, we have the triplet $J = 1$, with three possible projection $J_z = 0, \pm 1$. This gives the following matrix

$$H = \begin{bmatrix} -\epsilon & 0 & V \\ 0 & W & 0 \\ V & 0 & \epsilon \end{bmatrix}.$$

By solving the eigenvalue problem, the ground state energy can be exactly found.

The $J = 2$ case

Similarly, for the $N = 4$ particles case we have $J = 2$ with five possible projections $J_z = 0, \pm 1, \pm 2$. This gives us the Hamiltonian matrix

$$H = \begin{bmatrix} -2\epsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\epsilon + 3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \epsilon + 3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\epsilon \end{bmatrix}.$$

More rewriting

Following the work of LaRose and collaborators, see <https://journals.aps.org/prc/abstract/10.1103/PhysRevC.106.024319>, we can rewrite the Lipkin Hamiltonian for efficient adaptations to quantum computing.

We have

$$J_z = \sum_i^N j_z^i \quad J_{\pm} = \sum_i^N j_{\pm}^i = \sum_i^N (j_x^i \pm ij_y^i),$$

with N being the number of particles. Additionally, since we have spin-1/2 fermions, the mapping to Pauli matrices follow

$$j_x^i = \frac{1}{2}X_i, \quad j_y^i = \frac{1}{2}Y_i, \quad j_z^i = \frac{1}{2}Z_i.$$

This means that we require N qubits to calculate properties of a N particle system, if no more symmetry reductions are considered.

Lipkin Hamiltonian for quantum computing

We rewrote our Hamiltonian as

$$H_0 = \frac{\epsilon}{2} \sum_p Z_p,$$

$$H_1 = \frac{1}{2} V \sum_{p < q} (X_p X_q - Y_p Y_q),$$

$$H_2 = \frac{1}{2} W \sum_{p < q} (X_p X_q + Y_p Y_q).$$

For two particles coupling to spin $J = 1$ we have

$$H^{N=2} = \frac{\epsilon}{2} (Z_1 + Z_2) + \frac{W + V}{2} X_1 X_2 - \frac{W - V}{2} Y_1 Y_2.$$

For four fermions

Our Hamiltonian for $N = 4$ is

$$\begin{aligned} H^{N=4} = & \frac{\epsilon}{2} (Z_1 + Z_2 + Z_3 + Z_4) + \frac{W - V}{2} \left(X_1 X_2 + \right. \\ & X_1 X_3 + X_1 X_4 + X_2 X_3 + X_3 X_4 + X_3 X_4 \Big) \\ & + \frac{W + V}{2} \left(Y_1 Y_2 + Y_1 Y_3 + Y_1 Y_4 + Y_2 Y_3 \right. \\ & \left. + Y_3 Y_4 + Y_3 Y_4 \right). \end{aligned}$$

Note that well that a term like Z_1 reads $Z_1 \otimes I_2 \otimes I_3 \otimes I_4$ where the subscript points to qubit i and all matrices are 2×2 matrices.

Leaving out the W term

If we set $W = 0$ we can simplify the Hamiltonian. For this we note that if we only include the H_0 and H_1 terms, spins differing by ± 2 are the only possible non-diagonal couplings. Since H_0 is diagonal and single particle energies are degenerate, it does not break any symmetry. Instead of the $2J + 1$ spin projections, we simply have $J + 1$ relevant states. From <https://journals.aps.org/prc/abstract/10.1103/PhysRevC.106.024319>, the two-qubit Hamiltonian for $N = 4$ can be written as

$$H_{W=0}^{N=4} = \epsilon(Z_1 + Z_2) + \frac{\sqrt{6}}{2} V(X_1 + X_2 + Z_1 X_0 - X_1 Z_0).$$

This is computationally more efficient since we only need half of the qubits.

More details

Here we give some additional details behind the rewriting of the Lipkin Hamiltonian in term of Pauli matrices and the identity matrix. For the one-body term we have

$$H_0 = \epsilon J_z = \epsilon \sum_p j_z^p = \frac{\epsilon}{2} \sum_p Z_p$$

Moving on to the H_1 term, we need to expand the square of the $J_{\pm}$ operators. Starting with J_+ we find

$$\begin{aligned} J_+^2 &= \left(\sum_p j_x^p + \imath j_y^p \right)^2 = \sum_{pq} (j_x^p + \imath j_y^p)(j_x^q + \imath j_y^q) \\ &= \sum_{pq} (j_x^p j_x^q - j_y^p j_y^q + \imath j_y^p j_x^q + \imath j_x^p j_y^q) . \end{aligned}$$

Other manipulations

Similarly, the J_-^2 term yields

$$J_-^2 = \sum_{pq} (j_x^p j_x^q - j_y^p j_y^q - j_z^p j_x^q - j_z^p j_y^q).$$

Rewriting we get

$$\begin{aligned} J_+^2 + J_-^2 &= 2 \sum_{pq} j_x^p j_x^q - j_z^p j_z^q \\ &= 2 \sum_p (j_x^p)^2 - (j_y^p)^2 + 4 \sum_{p>q} (j_x^p j_x^q - j_y^p j_y^q) \\ &= \frac{1}{2} \sum_p (X_p^2 - Y_p^2) + \sum_{p>q} (X_p X_q - Y_p Y_q) \end{aligned}$$

Final expression for H_1 and H_2

The diagonal term will cancel out, since the Pauli matrices are involutory, resulting in

$$H_1 = \frac{1}{2}V \sum_{p>q} (X_p X_q - Y_p Y_q).$$

Following the same procedure as above, we have finally

$$H_2 = \frac{1}{2}W \sum_{p<q} (X_p X_q + Y_p Y_q).$$

Quantum computing and solving the eigenvalue problem for the Lipkin model

We turn now to a simpler variant of the Lipkin model without the W -term and a total spin of $J = 1$ only as maximum value of the spin. This corresponds to a system with $N = 2$ particles (fermions in our case). Our Hamiltonian is given by the quasispin operators (see below)

$$\hat{H} = \epsilon \hat{J}_z - \frac{1}{2} V (\hat{J}_+ \hat{J}_+ + \hat{J}_- \hat{J}_-).$$

As discussed earlier the quasispin operators act like lowering and raising angular momentum operators.

With these properties we can calculate the Hamiltonian matrix for the Lipkin model by computing the various matrix elements

$$\langle JJ_z | H | JJ'_z \rangle, \quad (17)$$

where the non-zero elements are given by

$$\begin{aligned} \langle JJ_z | H | JJ'_z \rangle &= \epsilon J_z \\ \langle JJ_z | H | JJ'_z \pm 2 \rangle &= \langle JJ_z \pm 2 | H | JJ'_z \rangle \\ &= \frac{1}{2} V \end{aligned}$$

Plans for next week

1. We will focus mainly on setting up the quantum circuit for the Lipkin model with $J = 1$ and $J = 2$, rewriting the Lipkin model in terms of Pauli matrices
2. Thereafter, if we get time we start with quantum Fourier transforms and discussions of Quantum Phase estimation algorithm